Homework 3: Coding Questions 7 and 11

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Question 7: HP Filter Kernel Matrix

The Hodrick–Prescott (HP) filter can be written in matrix form as

$$\hat{\theta} = (I + \lambda D^\top D)^{-1} y,$$

where D is the second-difference matrix and λ is the smoothing parameter. Defining

$$K = (I + \lambda D^\top D)^{-1},$$

we can write each fitted value as

$$\hat{\theta}_i = \sum_{j=1}^{n} K_{ij} y_j.$$

Thus, the i-th row of K describes how much each observed value y_j contributes to the smoothed value $\hat{\theta}_i$.

For this question, I used $n = 100$ and $\lambda = 100$. I computed the matrix K empirically and plotted rows $i = 25$, $i = 50$, and $i = 75$.

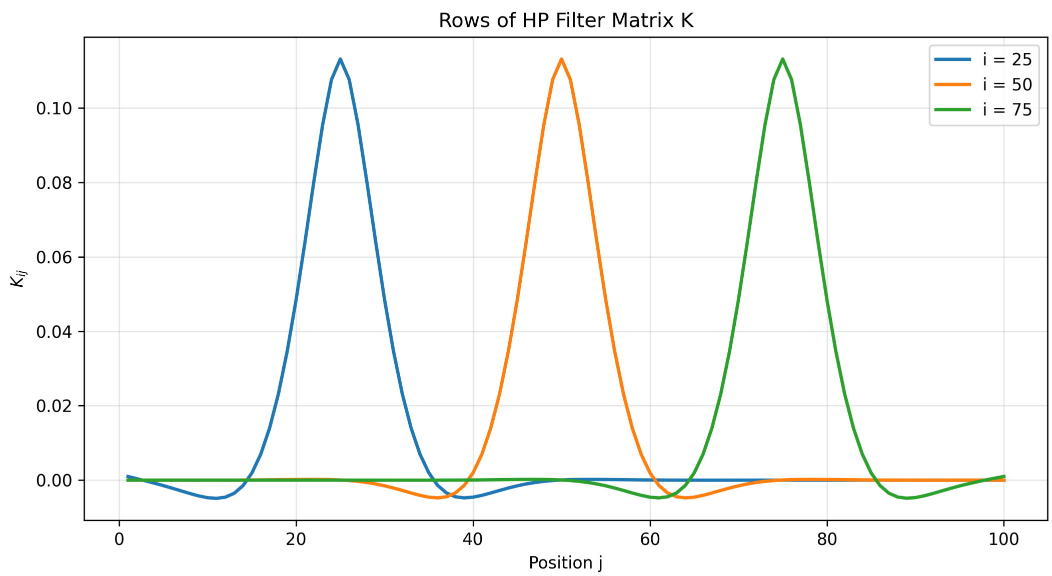

Figure 1: Rows 25, 50, and 75 of the HP filter smoothing matrix K.

Discussion

The plotted rows of K show how the HP filter behaves locally around each index i. Each curve has its largest weight near its own index and then gradually decays as we move away from that point. For example, the row corresponding to $i = 50$ places the most weight on observations close to time point 50, while observations farther away receive smaller weights.

This behaviour is similar to a kernel smoother. In a kernel smoother, the fitted value at a given point is computed as a weighted average of nearby observations, with nearby points receiving more weight than distant points. The HP filter behaves in the same way here: each $\hat{\theta}_i$ is a weighted average of the full data vector y, but the strongest weights are concentrated around i.

The curves for $i = 25$, $i = 50$, and $i = 75$ have similar shapes, except shifted to be centered around their respective indices. This supports the interpretation that the HP filter acts like a smoothing operation that uses local information from neighbouring time points.

Near the interior of the series, the kernel-like shape is fairly symmetric. Near the boundaries, one would expect more distortion because there are fewer observations available on one side. Since the selected rows are away from the endpoints, the kernel-like behaviour is easier to see.

Question 11: Empirical Verification of the Spectral Autocovariance Formula

In Question 10, the process is defined as

$$x_t = \sum_{j=1}^{p} \left(U_{j1} \cos(2\pi\omega_j t) + U_{j2} \sin(2\pi\omega_j t) \right),$$

where U_{j1} and U_{j2} are uncorrelated random variables with mean zero and variance σ_j^2. The theoretical autocovariance function for this process is

$$\gamma(h) = \sum_{j=1}^{p} \sigma_j^2 \cos(2\pi\omega_j h).$$

The corresponding theoretical autocorrelation function is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}.$$

For the empirical example, I generated a process with two frequency components:

$$\omega_1 = 0.05, \qquad \omega_2 = 0.15,$$

with variances

$$\sigma_1^2 = 4, \qquad \sigma_2^2 = 1.$$

I then computed the empirical autocorrelation function using the simulated time series and compared it with the analytic autocorrelation function derived from the formula above.

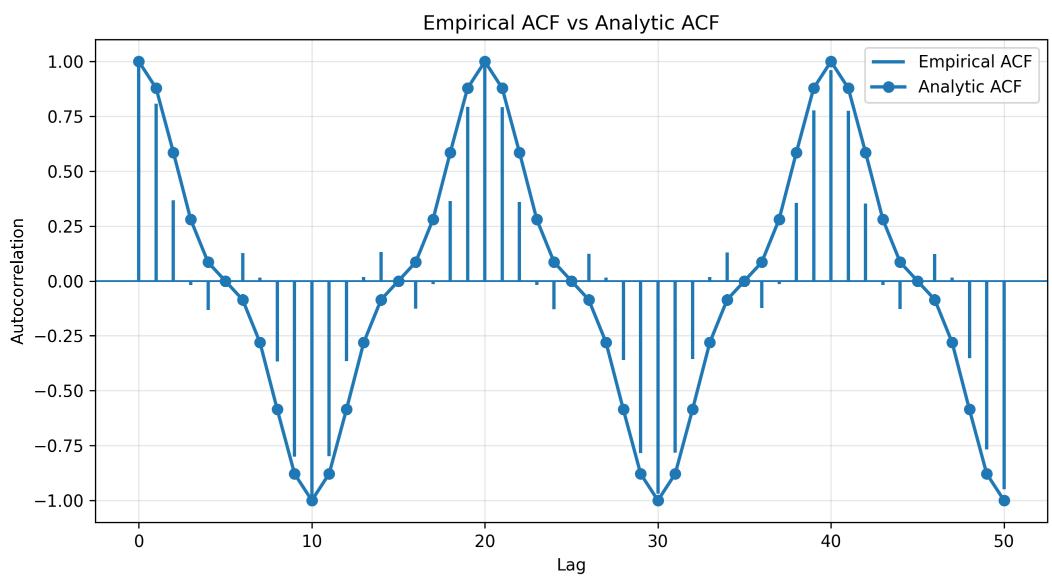

Figure 2: Empirical autocorrelation function compared with the analytic autocorrelation function.

Discussion

The empirical autocorrelation function shows an oscillatory pattern. This is expected because the process is constructed from sine and cosine waves. Since there are two frequency components, the autocorrelation is a combination of two cosine terms.

The analytic autocorrelation function also oscillates, following the formula

$$\rho(h) = \frac{\sum_{j=1}^{p} \sigma_j^2 \cos(2\pi\omega_j h)}{\sum_{j=1}^{p} \sigma_j^2}.$$

The empirical and analytic autocorrelation curves should have broadly similar patterns. They both show repeating cycles across lags, which reflects the periodic structure of the simulated process.

There may be small differences between the empirical and analytic curves because the empirical ACF is calculated from one finite simulated time series. With a longer time series or repeated simulations, the empirical autocorrelation would generally align more closely with the analytic formula.

Overall, this example verifies the theoretical autocovariance formula from Question 10. The process has an autocorrelation structure determined by the frequencies ω_j and the variances σ_j^2, and the empirical ACF reflects this predicted periodic behaviour.

Conclusion

For Question 7, the rows of the HP filter matrix K show that each fitted value is a weighted average of nearby observations. This supports the interpretation of the HP filter as a kernel-like smoother.

For Question 11, the simulated spectral process produced an empirical autocorrelation function that matches the oscillatory structure predicted by the analytic autocovariance formula. This provides an empirical check of the theoretical result from Question 10.